For a big tt-category whose compacts are local, reducedness of the fibre of the comparison map over zero is equivalent to a nilpotence alternative for two composable maps of compacts with vanishing tensor product

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Editorial note. The counted unit is the author's labelled theorem Thm:hom-local at source lines 881-893 with the proof environment that begins two lines later at source line 895; the pairing is the ordinary adjacent one.

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Editorial note. The counted proof refers only to its own two items and to the nilpotence theorem of the author's earlier work, which is cited and carried by the retained bibliography; every other label it uses is defined inside the retained statement, proof or context spans, so no reference is left to an external pointer.

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Allowed standard prerequisites (editorial).

- Triangular and homological spectra of a big tt-category. The notions of homological prime, of the comparison map phi to the triangular spectrum, and of the residue tensor functor are recalled in the delivered context. The equivalence proved in the counted theorem is the retained obligation.

- Local tt-categories and rigidity. Only the standing meaning of local (a tensor product of two non-zero compacts is non-zero) and the standard adjunction tricks using rigidity are assumed; the construction of the Serre tensor-ideal of modules killed by some non-zero compact is retained.
- Gabriel quotients, Serre tensor-ideals and flatness in locally coherent Grothendieck categories. Standard facts about Gabriel localization and about flat objects in the quotient are assumed; every step of the two implications is retained.

A.1 Theorem. *Let $\mathcal{T}$ be a big tt-category as in Equation* (A.2) *and assume its subcategory of rigid-compact $\mathcal{T}^c$ is* local*, meaning that $c \otimes d = 0$ with $c, d \in \mathcal{T}^c$ forces $c = 0$ or $d = 0$. Then the following are equivalent:*

i. The fiber of $\phi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \twoheadrightarrow \operatorname{Spc}(\mathcal{T}^c)$ over (0) is reduced to a single point.

ii. Whenever two maps f, g in $\mathcal{T}^c$ satisfy $f \otimes g = 0$ then either f is $\otimes$-nilpotent on some non-zero compact (that is, there exists $z \in \mathcal{T}^c$, $z \neq 0$ such that $f^{\otimes n} \otimes z = 0$ for $n \gg 1$) or g is $\otimes$-nilpotent on some non-zero compact.

Proof. Suppose (i) and let us prove (ii). By standard adjunction tricks using rigidity, we can assume that $f\colon \mathbb{1} \to x$ and $g\colon \mathbb{1} \to y$ both start from $\mathbb{1}$. Let $\mathcal{A} = \text{Mod-}\mathcal{T}^c$ as before and consider the subcategory of $\mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ given as follows

$$\mathcal{B}_0 := \big\{ M \in \mathcal{A}^{\mathrm{fp}} \,\big|\, \text{there exists } z \in \mathcal{T}^c,\ z \neq 0 \text{ with } \hat{z} \otimes M = 0 \big\}.$$

Since $\mathcal{T}^c$ is local, $\mathcal{B}_0$ is closed under extensions, hence $\mathcal{B}_0$ is a Serre $\otimes$-ideal of $\mathcal{A}^{\mathrm{fp}}$. Consider the Gabriel quotient

$$Q\colon \mathcal{A}^{\mathrm{fp}} \twoheadrightarrow \tilde{\mathcal{A}}^{\mathrm{fp}} := \mathcal{A}^{\mathrm{fp}}/\mathcal{B}_0.$$

Let us write $\tilde{z} \in \tilde{\mathcal{A}}^{\mathrm{fp}}$ for $Q(\hat{z})$ for all $z \in \mathcal{T}^c$. The maximal $\otimes$-ideals of $\tilde{\mathcal{A}}^{\mathrm{fp}}$ are in one-to-one correspondence with those of $\mathcal{A}^{\mathrm{fp}}$ containing $\mathcal{B}_0$. We claim that these are exactly those in $\phi^{-1}(0)$. Indeed, the kernel of the unit $\mathrm{coev}\colon \mathbb{1} \to \hat{z}^\vee \otimes \hat{z}$ in $\mathcal{A}^{\mathrm{fp}}$, being killed by $\hat{z}$ (in $\mathcal{T}^c$ already), becomes zero in $\tilde{\mathcal{A}}^{\mathrm{fp}}$ whenever $z \neq 0$. In other words, $\tilde{\mathbb{1}}$ is a subobject of $\tilde{z}^\vee \otimes \tilde{z}$ for every $z \in \mathcal{T}^c$ non-zero and in particular no proper $\otimes$-ideal of $\tilde{\mathcal{A}}^{\mathrm{fp}}$ contains any $\tilde{z}$ for $z \neq 0$. Hence the preimage $\phi(\mathcal{B}) = \mathbb{h}^{-1}(\mathcal{B}) = (0)$ is zero for every maximal $\otimes$-ideal of $\mathcal{A}^{\mathrm{fp}}$ containing $\mathcal{B}_0$. Under (i), we just proved that $\tilde{\mathcal{A}}^{\mathrm{fp}}$ has a unique maximal Serre $\otimes$-ideal, say $\mathcal{B}$. Consider now the kernels $M \rightarrowtail \tilde{\mathbb{1}} \xrightarrow{\tilde{f}} \tilde{x}$ and $N \rightarrowtail \tilde{\mathbb{1}} \xrightarrow{\tilde{g}} \tilde{y}$ in $\tilde{\mathcal{A}}^{\mathrm{fp}}$. Since $\tilde{f} \otimes M = 0$ and $\tilde{g} \otimes N = 0$ by the standard argument (using flatness of $\tilde{x}$ and $\tilde{y}$), we see that $\tilde{f}$ is $\otimes$-nilpotent on the Serre $\otimes$-ideal $\langle M \rangle$ generated by M and similarly for $\tilde{g}$ on $\langle N \rangle$. If, *ab absurdo*, both $\langle M \rangle$ and $\langle N \rangle$ are proper then they are both contained in the unique maximal Serre $\otimes$-ideal $\mathcal{B}$ of $\tilde{\mathcal{A}}^{\mathrm{fp}}$. Descending to $\tilde{\mathcal{A}}^{\mathrm{fp}}/\mathcal{B}$, we therefore have two monomorphisms $\bar{f}\colon \bar{\mathbb{1}} \rightarrowtail \bar{x}$ and $\bar{g}\colon \bar{\mathbb{1}} \rightarrowtail \bar{y}$, with $\bar{x}$ and $\bar{y}$ $\otimes$-flat. This implies that $\bar{f} \otimes \bar{g}\colon \bar{\mathbb{1}} \rightarrowtail \bar{x} \otimes \bar{y}$ is still a monomorphism. This contradicts the assumption that $f \otimes g = 0$. Consequently, one of the Serre $\otimes$-ideals $\langle M \rangle$ or $\langle N \rangle$ contains $\tilde{\mathbb{1}}$, hence $\tilde{f}$ or $\tilde{g}$ is $\otimes$-nilpotent in $\tilde{\mathcal{A}}^{\mathrm{fp}} = \mathcal{A}^{\mathrm{fp}}/\mathcal{B}_0$, and we conclude by definition of $\mathcal{B}_0$ that f or g is nilpotent on a non-zero compact.

Conversely, suppose (ii) and let us prove (i). Assume *ab absurdo*, that there exist two different homological primes $\mathcal{B}, \mathcal{C} \in \phi^{-1}((0))$ in the fiber of ϕ above zero. Then by [4, Prop. 5.3], we have $E_{\mathcal{B}} \otimes E_{\mathcal{C}} = 0$ and in particular the tensor of $\eta_{\mathcal{B}} \colon \mathbb{1} \to E_{\mathcal{B}}$ with $\eta_{\mathcal{C}} \colon \mathbb{1} \to E_{\mathcal{C}}$ is zero. It follows from the fact that $\hat{E}_{\mathcal{B}}$ and $\hat{E}_{\mathcal{C}}$ are colimits of representables and from finite presentation of $\hat{\mathbb{1}}$ that there exist factorizations $\eta_{\mathcal{B}} \colon \mathbb{1} \xrightarrow{f} x \to E_{\mathcal{B}}$ and $\eta_{\mathcal{C}} \colon \mathbb{1} \xrightarrow{g} y \to E_{\mathcal{C}}$ with $f \otimes g = 0$ and $x, y \in \mathcal{T}^c$. By (ii), one of f or g is $\otimes$-nilpotent on some non-zero compact, say f is. Hence there exists $z \neq 0$ in $\mathcal{T}^c$ such that $f^{\otimes n} \otimes z = 0$ for some $n \gg 1$. But then $\eta_{\mathcal{B}}^{\otimes n} \otimes z = 0$. Since $E_{\mathcal{B}}$ is a weak ring, $\eta_{\mathcal{B}} \otimes E_{\mathcal{B}}$ is a split monomorphism, hence $\eta_{\mathcal{B}}^{\otimes n} \otimes z = 0$ forces $E_{\mathcal{B}} \otimes z = 0$. The latter implies $0 \neq \hat{z} \in \mathcal{B}$, or $\phi(\mathcal{B}) = \mathbb{h}^{-1}(\mathcal{B}) \neq 0$, contradicting the choice of $\mathcal{B}$ in the fiber of (0). So the existence of two points in that fiber is absurd. □

A.2 Hypothesis. Let $\mathcal{T}$ be a *'big' tensor-triangulated category*, meaning a rigidly-compactly generated one, as in [7]. So $\mathcal{T}$ admits small coproducts; its subcategory $\mathcal{T}^c$ of compact objects coincides with that of rigid (strongly-dualizable) objects; $\mathcal{T}^c$ is essentially small and generates $\mathcal{T}$ as a localizing subcategory.

A.3 Definition. We say that an object A in $\mathcal{T}$ with a map $\eta \colon \mathbb{1} \to A$ (its 'unit') is a *weak ring* if $A \otimes \eta \colon A \to A \otimes A$ is a split monomorphism (whose retraction $A \otimes A \to A$ can be thought of as a unital non-associative multiplication on A).

A.4 Recollection. Let $\mathcal{C}$ in $\mathcal{A}$ be a localizing subcategory (i.e., closed under coproducts, extensions, subobjects and quotients). We have a Gabriel quotient $\mathcal{A}/\mathcal{C}$

$$\begin{array}{c} \mathcal{A} \\ Q_{\mathcal{C}} \big\downarrow \big\uparrow R_{\mathcal{C}} \\ \mathcal{A}/\mathcal{C} \end{array} \tag{A.5}$$

where $Q_{\mathcal{C}}$ is the universal exact functor with kernel $\mathcal{C}$, its right adjoint $R_{\mathcal{C}}$ is fully-faithful and $Q_{\mathcal{C}} \circ R_{\mathcal{C}} \cong \mathrm{Id}_{\mathcal{A}/\mathcal{C}}$. We shall only consider subcategories $\mathcal{C}$ that are $\otimes$-ideal ($\mathcal{A} \otimes \mathcal{C} \subseteq \mathcal{C}$), in which case $\mathcal{A}/\mathcal{C}$ inherits a unique tensor structure such that $Q_{\mathcal{C}}$ is a tensor functor. When $\mathcal{C}$ is clear from the context, we often write $\bar{X}$ instead of $Q_{\mathcal{C}}(\hat{X})$ for $X \in \mathcal{T}$. We also write $\bar{\mathcal{A}}_{\mathcal{C}}$ (or just $\bar{\mathcal{A}}$) for $\mathcal{A}/\mathcal{C}$ and

$$\bar{\mathbb{h}}_{\mathcal{C}} \colon \mathcal{T} \longrightarrow \bar{\mathcal{A}}_{\mathcal{C}} = \mathcal{A}/\mathcal{C} \tag{A.6}$$

for the composite $Q_{\mathcal{C}} \circ \mathbb{h}$, a coproduct-preserving cohomological tensor-functor.

A.7 Proposition. *With notation as in Equation* (A.4)*, we have:*

a. *The Gabriel quotient $\bar{\mathcal{A}} = \mathcal{A}/\mathcal{C}$ is a Grothendieck category, that is closed monoidal. Its tensor product preserves colimits in each variable.*

b. *Every $X \in \mathcal{T}$ has $\otimes$-flat image $\bar{X} = Q_{\mathcal{C}}(\hat{X})$ in $\bar{\mathcal{A}}$. If $x \in \mathcal{T}^c$ then $\bar{x}$ is rigid.*

c. *Every object of $\bar{\mathcal{A}}$ is a quotient of a coproduct $\bigsqcup_{i \in I} \bar{x}_i$ with all $x_i \in \mathcal{T}^c$.*

d. *Every injective object of $\bar{\mathcal{A}}$ is of the form $\bar{E}$ for some pure-injective E in $\mathcal{T}$, uniquely characterized by the property $\hat{E} \cong R_{\mathcal{C}}(\bar{E})$ in $\mathcal{A}$. In particular, $\bar{E}$ is $\otimes$-flat. Furthermore, $\bar{\mathbb{h}}_{\mathcal{C}} \colon \mathcal{T} \to \bar{\mathcal{A}}$ induces, for every $X \in \mathcal{T}$, an isomorphism*

$$\mathrm{Hom}_{\mathcal{T}}(X, E) \xrightarrow{\sim} \mathrm{Hom}_{\bar{\mathcal{A}}}(\bar{X}, \bar{E}).$$

e. *The right adjoint* $R_{\mathcal{C}}\colon \bar{\mathcal{A}} \to \mathcal{A}$ *is lax-monoidal and closed; more precisely for every* $M \in \mathcal{A}$ *and* $N \in \bar{\mathcal{A}}$*, if we denote by* $[-,-]$ *the internal-homs, we have*

$$[M, R_{\mathcal{C}}(N)]_{\mathcal{A}} \cong R_{\mathcal{C}}([Q_{\mathcal{C}}(M), N]_{\bar{\mathcal{A}}}).$$

f. *For every injective* $\bar{E}$ *in* $\bar{\mathcal{A}}$*, the internal-hom* $[-,\bar{E}]_{\bar{\mathcal{A}}}\colon \bar{\mathcal{A}}^{\mathrm{op}} \to \bar{\mathcal{A}}$ *is exact.*

Proof. These claims follow from [15, App. A] and [14, §2 & App. A]. Let us prove (f), for [14, Lem. 2.8] is only stated for $\mathcal{A}$. We have

$$[-,\bar{E}]_{\bar{\mathcal{A}}} \circ Q_{\mathcal{C}} = [Q_{\mathcal{C}}(-),\bar{E}]_{\bar{\mathcal{A}}} \cong Q_{\mathcal{C}} R_{\mathcal{C}}([Q_{\mathcal{C}}(-),\bar{E}]_{\bar{\mathcal{A}}}) \underset{\text{(e)}}{\cong} Q_{\mathcal{C}} \circ [-, R_{\mathcal{C}}(\bar{E})]_{\mathcal{A}}$$

and the latter is exact by [14, Lem. 2.8], since $R_{\mathcal{C}}\bar{E}$ is injective in $\mathcal{A}$ itself. As precomposition with $Q_{\mathcal{C}}$ detects exactness for functors on $\bar{\mathcal{A}}$, we get the result. □

A.8 Construction. For $\mathcal{C} \subseteq \mathcal{A}$ localizing $\otimes$-ideal, the injective hull of $\bar{\mathbb{1}}$ in $\bar{\mathcal{A}}_{\mathcal{C}}$ comes from $\bar{\mathbb{h}}_{\mathcal{C}}$, by Equation (A.7) (d). So there exists a morphism

$$\eta_{\mathcal{C}}\colon \mathbb{1} \to E_{\mathcal{C}}$$

in $\mathcal{T}$ such that $\bar{\eta}_{\mathcal{C}} = \bar{\mathbb{h}}_{\mathcal{C}}(\eta_{\mathcal{C}})\colon \bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ is the injective hull. It is characterized by $R_{\mathcal{C}}(\bar{E}_{\mathcal{C}}) \cong \hat{E}_{\mathcal{C}}$ in $\mathcal{A}$. Since $\bar{E}_{\mathcal{C}}$ is flat and injective, $\bar{E}_{\mathcal{C}} \otimes \bar{\eta}_{\mathcal{C}}$ is a split monomorphism in $\bar{\mathcal{A}}_{\mathcal{C}}$, hence by Equation (A.7) (d) again, the retraction exists in $\mathcal{T}$, meaning that $(E_{\mathcal{C}}, \eta_{\mathcal{C}})$ is a *weak ring* in $\mathcal{T}$ in the sense of Equation (A.3).

A.9 Recollection. A particular class of localizing $\otimes$-ideals $\mathcal{C} \subseteq \mathcal{A}$ are those generated by Serre $\otimes$-ideals $\mathcal{B} \subseteq \mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ of finitely presented objects (where $\otimes$-ideal only means $\mathcal{A}^{\mathrm{fp}} \otimes \mathcal{B} \subseteq \mathcal{B}$ of course). Explicitly $\mathcal{C} = \mathrm{Loc}(\mathcal{B})$ is the smallest localizing subcategory containing $\mathcal{B}$, which is then automatically $\otimes$-ideal in $\mathcal{A}$. In some notation, typically in indices, we drop the 'Loc' part and write only $\mathcal{B}$, like for instance with the canonical cohomological tensor-functor (A.6):

$$\bar{\mathbb{h}}_{\mathcal{B}}\colon \mathcal{T} \longrightarrow \bar{\mathcal{A}}_{\mathcal{B}} := \mathcal{A}/\operatorname{Loc}(\mathcal{B}). \tag{A.10}$$

Similarly, $E_{\mathcal{B}}$ means $E_{\mathrm{Loc}(\mathcal{B})}$ as in Equation (A.8) for $\mathcal{C} = \mathrm{Loc}(\mathcal{B})$.

A.11 Proposition. *With notation as in Equation* (A.9)*, we have:*

a. *The Grothendieck category* $\bar{\mathcal{A}}_{\mathcal{B}}$ *is still locally coherent. Its finitely presented objects* $\bar{\mathcal{A}}^{\mathrm{fp}} = \mathcal{A}^{\mathrm{fp}}/\mathcal{B}$ *coincide with the Gabriel quotient of* $\mathcal{A}^{\mathrm{fp}}$ *by* $\mathcal{B}$*.*

b. *We can recover* $\mathcal{B}$ *and* $\mathrm{Loc}(\mathcal{B})$ *from the pure-injective weak ring* $E_{\mathcal{B}}$ *of Equation* (A.8)*, as* $\mathcal{B} = \left\{ M \in \mathcal{A}^{\mathrm{fp}} \,\middle|\, \hat{E}_{\mathcal{B}} \otimes M = 0 \right\}$ *and* $\mathrm{Loc}(\mathcal{B}) = \operatorname{Ker}(\hat{E}_{\mathcal{B}} \otimes -)$*.*

c. *Every subobject of a finitely presented object in* $\bar{\mathcal{A}}_{\mathcal{B}}$ *is the colimit of its finitely presented subobjects. In particular, if* $I \rightarrowtail \bar{\mathbb{1}}$ *in* $\bar{\mathcal{A}}_{\mathcal{B}}$ *then* $I = \operatorname*{colim}_{f.p.\ M \rightarrowtail I} M$*.*

A.12 Recollection. The *homological spectrum* $\mathrm{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ consists of all maximal Serre $\otimes$-ideals $\mathcal{B}$ of the abelian subcategory $\mathcal{A}^{\mathrm{fp}} = \text{mod-}\mathcal{T}^c$ of finitely presented $\mathcal{T}^c$-modules. We call those $\mathcal{B}$ the *homological primes* of $\mathcal{T}$. Note that they only

depend on the subcategory $\mathcal{T}^c$. For each $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$, its preimage in $\mathcal{T}^c$ under Yoneda

$$\phi(\mathcal{B}) = \mathbb{h}^{-1}(\mathcal{B}) \cap \mathcal{T}^c = \big\{ x \in \mathcal{T}^c \,\big|\, \hat{x} \in \mathcal{B} \big\}$$

is a (triangular) prime ideal in $\mathcal{T}^c$. This defines a surjection $\phi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \to \to \operatorname{Spc}(\mathcal{T}^c)$ by [4, Cor. 3.9]. Each $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ yields a coproduct-preserving homological tensor-functor $\bar{\mathbb{h}}_{\mathcal{B}} \colon \mathcal{T} \to \bar{\mathcal{A}}_{\mathcal{B}} = \mathcal{A}/\operatorname{Loc}(\mathcal{B})$ as in (A.10).

A.13 Remark. The maximality of $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ among the Serre $\otimes$-ideals of $\mathcal{A}^{\mathrm{fp}}$ tells us that $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is 'simple': it has only the two trivial Serre $\otimes$-ideals, zero and $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$. (See Equation (A.9) if necessary.) Simplicity has the following consequence.

A.14 Proposition. *Let $\mathcal{B} \in \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c)$ and $A \in \bar{\mathcal{A}}_{\mathcal{B}}$ be a weak ring (Equation (A.3)) that is $\otimes$-flat, like $A = \bar{E}$ for some weak ring $E \in \mathcal{T}$. Then either $A = 0$ or its unit is a monomorphism $\bar{\mathbb{1}} \rightarrowtail A$.*

Proof. This is a basic trick in the proof of the Nilpotence Theorem [4]. Consider the exact sequence $\operatorname{Ker}(\eta) \rightarrowtail \bar{\mathbb{1}} \xrightarrow{\eta} A$ in $\bar{\mathcal{A}}_{\mathcal{B}}$. Since A is flat and $A \otimes \eta$ is a (split) monomorphism, we have $A \otimes \operatorname{Ker}(\eta) = 0$. Suppose that $\operatorname{Ker}(\eta) \neq 0$ and let us show that $A = 0$. By Equation (A.11) (c), we know that $\operatorname{Ker}(\eta)$ is the colimit of its finitely presented subobjects. Take a finitely presented $M \neq 0$ with $M \rightarrowtail \operatorname{Ker}(\eta)$. By flatness of A and $A \otimes \operatorname{Ker}(\eta) = 0$, we see that $A \otimes M = 0$. Hence $\operatorname{Ker}(A \otimes -) \cap \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is a non-zero Serre (by flatness) $\otimes$-ideal of $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$. As $\bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ is 'simple' (Equation (A.13)) this forces $\operatorname{Ker}(A \otimes -) \cap \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}} = \bar{\mathcal{A}}^{\mathrm{fp}}_{\mathcal{B}}$ to contain $\bar{\mathbb{1}}$, giving $A = 0$. □

We want to discuss the injectivity of the map $\phi\colon \operatorname{Spc}^{\mathrm{h}}(\mathcal{T}^c) \twoheadrightarrow \operatorname{Spc}(\mathcal{T}^c)$, from the homological spectrum (Equation (A.12)) to the usual triangular spectrum $\operatorname{Spc}(\mathcal{T}^c)$. By a general finite-smashing localization argument, we can reduce to the case where $(0) \in \operatorname{Spc}(\mathcal{T}^c)$, meaning that $\mathcal{T}^c$ is *local*. For instance, $\mathcal{T}$ could be $\mathrm{SH}_{(p)}$ the p-local stable homotopy category or $\mathrm{D}(R)$ the derived category of a commutative local ring R. Under this assumption, we have the following characterization:

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